CancerOS Literature Review and Implementation Addendum

Reference: COS-RA-001

Date: 23 October 2026

Status: Research recommendations for review

Scope: Seven supplied references compared with the drafted CancerOS specifications

1 Recommendation

Keep the established Python/FastAPI, Next.js, PostgreSQL, private-object-storage and worker architecture. Strengthen the prototype through visible evidence, measurement traceability and explicit limitations. The papers do not justify turning the hackathon into a diagnostic classifier, HER2 predictor or validated treatment-response forecasting system.

Prioritize three additions: an Evidence and limitations panel, a Why did this change view, and a controlled geometric sensitivity experiment. The first two consolidate behavior already required by the specifications. The sensitivity experiment is an optional new research demonstration and must not be described as calibrated clinical uncertainty. Preserve the five hypothetical families and at least one verified genuine longitudinal case.

The full specification set is already drafted. This addendum proposes focused implementation refinements, not another round of drafting the entire set. Drafted requirements are not evidence of completed or validated code. No repository implementation was audited for this review, and no source specification is silently revised.

2 Verification method and access limits

Checked supplied URLs, publisher/index records and accessible primary research text on 23 October 2026. Used original-study abstracts and institutional research accounts where full text was inaccessible. Several PMC requests encountered a browser challenge; MDPI and ScienceDirect returned access/rate-limit errors, and the SPIE PDF was not retrieved. Indexed publisher text supports bibliographic and abstract-level findings for those items, not a complete methods or risk-of-bias appraisal.

This is a targeted engineering literature assessment, not a new systematic review or a reproduction of reported experiments. Review articles supply an overview; their strongest individual scores cannot be transferred into CancerOS. The original studies address different endpoints, modalities and cohorts from the proposed longitudinal research prototype.

3 Paper verification ledger

R1 Broad imaging and explainability review

The linked Cancers article is by Margo Sabry and colleagues, not Cioni and colleagues; Dania Cioni is the academic editor. The publisher describes 65 studies from 2018–2024. Its headline figures include mammography accuracy up to 98.5 percent and histopathology ViT accuracy of 96 percent. These are different modality/task results, not a pooled MRI performance guarantee. Bibliography and indexed publisher/abstract text were verified; full-text methods appraisal was restricted by access. Useful engineering lesson: make explanation and source context visible, while preserving task-specific evaluation. [R1]

R2 Machine learning and deep learning systematic review

The fuller title is Machine learning and deep learning for breast cancer: A decade systematic review of detection, classification, prognosis, and explainability, by Owotogbe and colleagues, DOI 10.1016/j.imu.2026.101756. Publisher-

indexed abstract text confirms 131 studies from 2016–2025. Full-text inspection was unavailable. Use this as landscape context for multimodal integration and explainability, not as an independently verified model recipe or proof that fusion will improve this prototype. [R2]

R3 Management continuum review

The linked article is Artificial intelligence for breast cancer management by Chua and colleagues, published in Communications Medicine on 3 January 2026, not Nature Communications in 2025. Its scope spans multiple phases of management and discusses validation and generalizability. Metadata and the original article abstract were verified through PubMed and the authors' institution; publisher full text was inaccessible. It supports the research framing of integrated longitudinal information, not proof of a working patient-specific digital twin. [R3]

R4 MRI focused systematic review

The publisher lists Deep Learning Methods for Breast Cancer Detection, Classification, and Segmentation Using MRI Scans: A Systematic Review, by Qais Al-Azzam and colleagues, AI Med. 2026, 1(3), 24, dated 1 September 2026. Indexed text describes MRI studies from 2014–2025 and limitations involving external validation and acquisition variability. Full-text retrieval was unavailable. Its useful direction is to distinguish detection, classification and segmentation tasks and respect protocol compatibility; it does not establish a ready-to-run checkpoint. [R4]

R5 MRI and clinical HER2 prediction

Qiu and colleagues report 6,438 screened patients with 2,400 included and AUC 0.94. Crucially, the accessible full text states that the principal result is internally validated and independent external validation is needed. Other passages refer to cross-center/external performance, creating reporting ambiguity; an unqualified external-validation claim is unwarranted. Data are available on request, not automatically downloadable with reusable weights. Transfer the idea of separately evaluated imaging/clinical features, not the reported score or a deployed HER2 classifier. [R5]

R6 MRI diagnostic reader study

Witowski and colleagues' study is from 2022 in Science Translational Medicine, DOI 10.1126/scitranslmed.abo4802. The original abstract reports internal AUROC 0.92 on 3,936 examinations and retrospective reader/external-dataset evaluation. A nonsignificant reader difference does not by itself prove clinical equivalence. The biopsy-avoidance estimate is a retrospective decision-analysis result, not demonstrated prospective benefit from CancerOS. Adopt its emphasis on external testing and failure analysis. Original abstract/index record was reviewed; full PMC text was blocked. [R6]

R7 Uncertainty weighted learning

The October 2026 news links to Li, Arefan and Wu's SPIE work, DOI 10.1117/12.3087768. PSC's institutional account describes mammography-based uncertainty-weighted learning. The researchers' university poster abstract also describes risk-coverage evaluation and mapping uncertainty to prediction risk. The conference full paper was not retrieved, so its quantitative results and reproducibility are not fully appraised here. This is an early model-specific research method, not an uncertainty plug-in for breast MRI or an LLM confidence score. [R7]

4 What the literature means for the established specifications

Area	Existing CancerOS decision	Recommended action
Imaging, COS-SPEC-005/007	Verified imported masks, physical measurements and optional evaluated segmentation	Retain. Add a visible image/mask/measurement explanation path.
Fusion, COS-SPEC-006/007	Structured late fusion with modality availability	Retain. Add explicit contribution and missingness sections; do not train a large fusion network for P0.
Persistence, COS-SPEC-008	Provenance edges, immutable snapshots, model registry and exact claim links	Reuse. Most new views can project existing records.
Interface, COS-SPEC-009	Findings, Records and Evidence panels; distinct origins	Consolidate into a coherent Evidence and limitations experience, not another crowded dashboard.

Area	Existing CancerOS decision	Recommended action
Integrity, COS-SPEC-010	No invented confidence, HER2 or treatment claim; source/synthetic separation	Retain. Distinguish geometric stress tests from predictive uncertainty.
Evaluation, COS-SPEC-011	Patient-group splits, optional-model gates, meaningful baselines and failure reporting	Retain. Add explicit no-transfer of literature scores and optional risk-coverage evaluation.
Delivery, COS-SPEC-012	Real-data vertical slice before advanced models	Retain. Add small tasks after the corresponding core capability, not a new architecture.

The papers largely reinforce existing decisions. The remaining opportunity is making those decisions obvious in the working demonstration. Multimodal means linked verified information from more than one modality; it does not require every patient to have all modalities or a trained end-to-end neural network.

5 Recommended feature one Evidence and limitations panel

For each selected snapshot, show five independent sections: source linkage, annotation/geometry quality, available modalities, evidence support and analysis limitations. Use states such as checked, warning, unavailable and not evaluated, with the named rule and evidence reference. Never combine them into an arbitrary 0–100 trust score or call a passed engineering check clinically validated.

Implementation:

1. Resolve the authorized state hash and exact member IDs.
2. Aggregate existing QC, provenance, modality and claim-validation records with deterministic rules.
3. Return reasons and source references, not model-authored confidence labels.
4. Render the result in the contextual inspector; selecting a reason opens its authorized record.
5. Bind responses to patient/state/branch selection to prevent stale-result leakage.

Example: source linkage reviewed; anatomical-mask QC passed; no laboratory observations recorded; background literature available; individual treatment response not evaluated. These are different facts and must remain separate.

Reuse state modalities, imaging QC, validation results, provenance and claim-link tables. A computed response can initially be assembled from existing authorized reads. If a consolidated endpoint is added, propose GET /patients/{p}/states/{s}/evidence-status under the project API base, version its schema and add authorization tests. This route is a proposed amendment, not an existing COS-SPEC-008 capability.

Acceptance: missing laboratories remain missing; a functional analysis mask cannot acquire an anatomical-contour pass; an inaccessible source stays inaccessible; no probabilistic trust percentage is emitted. This is the highest-value low-compute addition.

6 Recommended feature two Why did this change view

Explain an observed comparison through a deterministic chain: source examinations and masks, physical-space measurements, compatible lesion definition, absolute/percentage change and applicable limitations. Clicking a chart difference should open this chain rather than a generic AI paragraph.

Reuse comparison endpoints and provenance/member links. The server supplies metric, units, from/to IDs, comparability outcome and reason codes. The UI displays real or hypothetical context and exact source times. No Grad-CAM is necessary for a deterministic volume calculation; show the mask overlay and method instead. Do not create a model saliency heatmap when no corresponding evaluated classifier exists.

For any later enabled model, an attribution explanation must identify the specific model, target, input and method, and undergo stability/fidelity checks. A visually plausible explanation is not proof of correct prediction or causal biology. Do not portray transformer attention as a validated explanation merely because it is available.

Acceptance: incompatible annotation classes yield not_comparable; zero baseline yields a missing percentage with a reason; source dates are not invented; the same values appear in chart, record panel and explanation. A changed source revision creates a new explanation/snapshot reference, preserving historical results.

7 Recommended feature three Controlled mask sensitivity experiment

Offer a clearly labeled engineering experiment showing how a measurement depends on an assumed boundary. Starting from an approved mask, create a few controlled physical-distance erosions/dilations or boundary variations, measure each and display their volumes beside the baseline. Use physical spacing, not arbitrary equal voxel counts on anisotropic data. Keep this out of the observed clinical timeline.

The displayed min/max is a geometric perturbation range, not a confidence interval, annotation uncertainty estimate or plausible patient future. Fixed offsets do not establish how an expert would draw the boundary. A narrow range does not prove accuracy. This restriction is essential: the feature borrows the principle of examining uncertainty without claiming to reproduce UWL.

Execution contract: require source segmentation ID, baseline hash, algorithm/configuration version, offset units and requested offsets. Worker outputs are a private experiment artifact with exact variant measurements, failed/empty variants and lineage. They do not create new patient observations or studies. Quarantine unsupported masks and failed geometry. Explicitly label variants artificial boundary perturbations; assign a distinct experimental resource kind rather than converting them to REAL_OBSERVED or patient predictions.

Add a versioned experiment catalog/result schema and authorized job endpoints only if this optional feature is selected. Proposed routes are POST /patients/{p}/measurement-sensitivity-jobs and GET /patients/{p}/measurement-sensitivity/{id}. Patient/project ownership, worker leases, idempotency and artifact grants follow COS-SPEC-008. These additions require review; no new production feature is authorized by a paper's existence alone.

Acceptance: millimetre offsets behave correctly on anisotropic fixtures; zero/empty cases are reported; source content is unchanged; experiment results never enter clinical or prediction metrics; CPU/RAM limits are measured. Precompute approved results remotely if the laptop cannot process them reliably.

8 Conditional feature DCE examination explorer

If the selected source actually supplies compatible phases and timing, allow navigation of pre/post-contrast acquisitions within one examination. A lesion-region signal curve can be a useful descriptive research view. It must remain separate from the cross-visit tumor-volume timeline.

Require verified phase order, source acquisition times or explicitly ordinal labels, aligned ROI, channel semantics, and preprocessing provenance. If an enhancement ratio is computed, define its formula and handle zero/near-zero baseline signal explicitly. Motion, scanner settings and normalization can alter intensity curves. Do not infer HER2, perfusion, resistance or treatment benefit from the curve. Do not impose a diagnostic washout label without an implemented reviewed definition.

This is P1: it can strengthen the imaging explanation but is less valuable than a reliable source-linked lesion and genuine longitudinal case. When only a processed volume is available, show the supported volume without inventing missing phases.

9 Later research rather than hackathon requirements

Research direction	Necessary work before enablement
HER2 prediction	Verified labels, compatible sequences, independent original-patient splits, available licensed model/code, external evaluation and calibration; source pathology status stays separate
Frozen image embeddings plus simple classifier	Predefined endpoint, training-only feature selection and patient-held-out comparisons with a simple baseline; no diagnosis endpoint by default

Research direction	Necessary work before enablement
UWL or other learned uncertainty	Reproducible method, suitable labeled MRI task, calibration data and held-out testing; uncertainty scores are not calibrated probabilities by assumption
Risk-coverage analysis	Actual model errors and uncertainty ranking on a frozen set; plot retained-case coverage against error, report excluded cases and subgroup effects
Predictive digital twin	Sufficient genuine temporal outcomes, defined horizons, time-aware baselines and independent calibration; synthetic branches are not validation data

For any optional classifier, AUC assesses ranking, not threshold accuracy or calibration. Report sensitivity/specificity at a predeclared threshold, AUROC/AUPRC where appropriate, calibration and all failures. If no classification endpoint is enabled, these metrics are not applicable; use the geometry, integrity and grounding metrics in COS-SPEC-011 instead.

The supplied work does not establish that a small curated cohort can support individualized prognosis or treatment optimization. It also does not demonstrate that an LLM can fill missing pathology labels. Keep unknown HER2 unknown in the source record even if a future research model estimates it separately.

10 Implementation sequence and effort assumptions

These tasks assume the existing real-data vertical slice is working. They do not replace B01–B30 or imply that those tasks are already implemented. Estimates are provisional focused developer effort, excluding source review and integration failures.

Task	Work and dependencies	Target effort
RA01	Correct literature metadata and build evidence cards with design/endpoint/cohort/validation/access limits; after B20	Half day
RA02	Define evidence-status schema and deterministic rule mapping; after B06/B10 and QC availability	Half day
RA03	Build Evidence and limitations panel with source drill-down; after RA02/B23	Half to one day
RA04	Add Why did this change to comparisons with mask/method links; after B08/B11/B12	Half to one day
RA05	Implement controlled perturbation worker/result type and physical tests; after B08/B14; optional	One to two days
RA06	Freeze new regression cases, review wording and rehearse; after selected RA tasks	Half day
RA07	Add DCE examination explorer only if verified phases and time remain	Separate estimate after data inspection

Do RA01–RA04 before RA05. If the deadline is tight, deliver the first two views only. Never sacrifice five scenario families, real follow-up verification, permission checks or source-versus-simulation labels for a new experimental feature. Do not begin a new large CNN/ViT training effort on the 8 GB laptop. Remote preprocessing and compact prepared assets remain the practical path.

11 Evaluation additions

Add tests for evidence-status derivation, source/branch ownership, stale response rejection, contradictory QC inputs, missing modalities and no invented trust scores. Extend comparison tests with source-provenance drill-down and exact agreement between rendered and backend values. If the perturbation experiment is built, test physical offsets, reproducible outputs, empty-mask behavior, artifact permissions and separation from patient states.

No new clinical performance threshold is proposed. Publish actual test counts and failure examples. A binary QC rule's pass rate is not model calibration. If risk-coverage evaluation is later added, include both error and retained coverage so abstaining on most inputs cannot masquerade as universally reliable prediction.

12 Demonstration refinement

Keep the existing five-minute presentation structure. Within the real baseline step, open Why did this change and show its exact source/method. Within the findings step, open Evidence and limitations and show one missing-modality or incompatibility reason. If RA05 is ready, replace part of an optional scenario flourish with a short geometric perturbation demonstration, explicitly labeled artificial and nonclinical.

Suggested project statement: “CancerOS links observed breast cancer research data, reproducible hypothetical trajectories and evidence-backed explanations in an interactive longitudinal workspace. It makes the limits of each result inspectable.” This is a proposed positioning statement, not a claim of novelty priority or clinical approval.

The most persuasive technical sequence is real image/mask, derived lesion and measurement, real follow-up comparison, clearly hypothetical branch, then source-linked explanation and abstention where evidence is insufficient. A single uncertainty percentage or a copied literature accuracy headline would weaken that evidence.

13 Specification amendments for review

Specification	Proposed focused change
COS-SPEC-005	Add measurement explanation inputs; optional experimental boundary perturbations and DCE eligibility checks
COS-SPEC-006	Require structured explanation facets; keep learned uncertainty and HER2 estimation separately gated
COS-SPEC-007	Add a small optional experiment worker module; no new infrastructure platform
COS-SPEC-008	Define computed evidence-status response; optional experiment result resource/jobs; preserve ownership and immutability
COS-SPEC-009	Refine the existing inspector with Evidence and limitations and Why did this change; avoid duplicate panels
COS-SPEC-010	Explicitly prohibit labeling artificial boundary variation as calibrated medical uncertainty
COS-SPEC-011	Add explanation/experiment regressions; reserve risk-coverage/calibration for an actual evaluated prediction task
COS-SPEC-012	Add RA01–RA06 behind existing core tasks with a clear optional-feature cutoff

Draft changes should be reviewed for consistency before implementation. Do not change original requirements solely to match an attractive paper result. Full model, data and license due diligence remains necessary before code or weights from any paper are adopted.

14 Verified references and access record

- R1 Sabry and colleagues. AI-Driven Breast Cancer Diagnosis: A Systematic Review of Imaging Modalities, Deep Learning, and Explainability. Cancers 18(8), 1305, 2026. Publisher/indexed abstract verification; direct full text blocked. https://www.mdpi.com/2072-6694/18/8/1305 and https://pmc.ncbi.nlm.nih.gov/articles/PMC13114346/
- R2 Owotogbe and colleagues. Machine learning and deep learning for breast cancer: A decade systematic review of detection, classification, prognosis, and explainability. Informatics in Medicine Unlocked, 2026. Publisher-indexed abstract; full text inaccessible. https://doi.org/10.1016/j.imu.2026.101756
- R3 Chua and colleagues. Artificial intelligence for breast cancer management. Communications Medicine 6, 79, 2026. Original abstract/metadata and author-institution page; publisher full text blocked. https://pubmed.ncbi.nlm.nih.gov/41484464/ and https://www.nature.com/articles/s43856-025-01342-3

- R4 Al-Azzam and colleagues. Deep Learning Methods for Breast Cancer Detection, Classification, and Segmentation Using MRI Scans: A Systematic Review. AI Med. 1(3), 24, 2026. Indexed publisher material; full text unavailable. https://www.mdpi.com/3042-6707/1/3/24
- R5 Qiu, Zhao and Zhao. Novel deep learning-based prediction of HER2 expression in breast cancer using multimodal MRI, nomogram, and decision curve analysis. Frontiers in Oncology, 2025. Accessible full text inspected for validation and data-availability qualifications. https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2025.1593033/full
- R6 Witowski and colleagues. Improving breast cancer diagnostics with deep learning for MRI. Science Translational Medicine 14(664), eabo4802, 2022. Original abstract/index record and code-availability search record; full PMC retrieval blocked. https://pubmed.ncbi.nlm.nih.gov/36170446/ and https://doi.org/10.1126/scitranslmed.abo4802
- R7 Li, Arefan and Wu. Uncertainty-weighted learning to reduce diagnostic uncertainty, evaluate AI models, and augment precise clinical decisions. SPIE, 2026, DOI 10.1117/12.3087768. Full conference paper unavailable; institutional research account and researchers' university poster abstract reviewed. https://www.psc.edu/breast-cancer-diagnosis/ and https://www.aihub.pitt.edu/news/data-science-and-ai-forum-day-one-research-poster-showcase

Project comparison used current Library content from COS-SPEC-007, COS-SPEC-008, COS-SPEC-009, COS-SPEC-010 and COS-SPEC-011. Earlier simulation/intelligence scope and the established delivery backlog were available in the current drafting context. Recommendations are engineering judgments informed by this literature, not experimentally validated CancerOS results.